What Did You Learn?

23.1 Ancestry and Features of Land Plants

Land **plants** evolved from a common ancestor with multicellular, freshwater algae about 590 MYA. The **charophytes** *Chara* and *Coleochaete* are green algae related to the ancestors of plants and, like other algae, have characteristics in common with plants, such as photosynthetic pigments, the presence of starch, and cell wall structure.

- Plants evolved modifications for life on land, such as a waxy **cuticle** to combat desiccation and **stomata** for gas exchange.
- Specialized cells called **tracheids** facilitate the movement of water within a plant, and the formation of tissues and organs reduces surface water loss. The **alternation-of-generations** life cycle contains a multicellular diploid stage that can be beneficial if mutations are caused by UV light.
- The **sporophyte** is the diploid portion of the life cycle; here, specialized structures called **sporangia** produce haploid spores. **Spores** divide by mitosis and become the gametophyte.
- The haploid stage is the multicellular **gametophyte,** with male and female gametangia called the **antheridia** and **archegonia.** Sperm and egg are produced, fuse, and form a diploid zygote.

23.2 Evolution of Bryophytes: Colonization of Land

Ancient **bryophytes** were the first plants to colonize land. **Liverworts, hornworts,** and **mosses** are examples of bryophytes, which lack well-developed **vascular tissue** and are therefore called **nonvascular plants.**

- Without true roots, **rhizoids** instead anchor these plants to a substrate. The sporophyte is nutritionally dependent on the gametophyte, because the gametophyte is larger and photosynthetic.
- The life cycle of mosses demonstrates the reproductive strategies of flagellated sperm and dispersal by means of windblown spores.

23.3 Evolution of Lycophytes: Vascular Tissue

Vascular plants, such as rhyniophytes, evolved during the Silurian period.

- The sporophytes contain two types of conducting tissues: **Phloem** is specialized to move organic nutrients, whereas **xylem** is specialized to conduct water and dissolved minerals with the help of **lignin** in xylem cells.
- **Lycophytes** are descended from these first plants. They contain vascular tissue, produce spores, and are called the **seedless vascular plants.**
- Ancient lycophytes were the first plants to have leaves known as **microphylls,** and modern lycophytes have conelike structures called **strobili,** where sporangia form. An underground stem called a **rhizome** has extensions of roots.
- Seedless vascular plants can produce one type of spore, and be **homosporous,** or two types of spores, and be **heterosporous.** The two types of spores made by hetersporous plants are **microspores** and **megaspores.**

23.4 Evolution of Pteridophytes: Megaphylls

In the **pteridophytes** (**ferns** and their allies, **horsetails** and **whisk ferns**) and lycophytes, the sporophyte is the dominant stage of the life cycle and is separate from the tiny gametophyte.

- Windblown spores are the dispersal agents for these plants. The ferns found on Earth today have an obvious **megaphyll;** horsetails and whisk ferns have reduced megaphylls.
- Fern megaphylls are called **fronds,** and there is much diversity in the shape and size of these fronds. Most fronds have sporangia on the underside, found in clusters called **sori.**
- Established ferns can often reproduce asexually by growing new plants from the underground rhizome.

23.5 Evolution of Seed Plants: Full Adaptation to Land

Seed plants also have an alternation-of-generations life cycle, but they are heterosporous, producing both microspores and megaspores.

- Microspores become the windblown or animal-transported male gametophytes—the **pollen grains. Pollination** involves pollen grains carrying and delivering sperm via a **pollen tube** to the female gametophyte (megaspore), which is the **ovule.** Following fertilization, the ovule becomes the **seed,** which contains a sporophyte embryo. Fertilization no longer requires external water, and sexual reproduction is fully adapted to the terrestrial environment.
- The **gymnosperms** (cone-bearing plants) evolved from woody seed ferns during the Devonian period. The **conifers,** represented by the pine tree, exemplify the traits of these plants. Other gymnosperms are the **cycads, ginkgoes,** and **gnetophytes.** Gymnosperms have "naked seeds," because they are not enclosed by fruit, and can be **monoecious** (having both male and female structures on the same plant) or **dioecious** (being either male or female but not both).
- *Nanjinganthus dendrostyla* has been identified as most closely related to the common ancestor for the angiosperms. **Angiosperms** belong to two classes: **Monocotyledones (monocots)** with one **cotyledon,** and **Eudicotyledones (eudicots)** with two cotyledons. The reproductive organs are found in **flowers.** Most flowers have **sepals, petals, stamens,** and **carpels.** After fertilization, the ovules become seeds, which are located in the **ovary.** Seeds are the result of **double fertilization,** with two sperm yielding the embryo, and the endosperm to feed the embryo. This ultimately becomes the **fruit.** Therefore, angiosperms have "covered seeds."
- In many angiosperms, pollen is transported from flower to flower by various pollinators. Both flowers and fruits are found only in angiosperms and may account for the extensive colonization of terrestrial environments by the flowering plants.

Thinking Critically About Biology

1. For each plant listed next, choose all the plant features (a.–j.) that can be used to describe that plant.

Pine tree	a. vascular tissue
Moss	b. no vascular tissue
Apple tree	c. protected embryo
Fern	d. megaphylls
Horsetail	e. gametophyte is the dominant generation
	f. produces seeds
	g. produces fruit
	h. alternation-of-generations life cycle
	i. disperses spores
	j. sporophyte is the dominant generation

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